

EA Credit: Renewable Energy

This prerequisite applies to

- ▶ BD+C: New Construction (1-5 points)
- ▶ BD+C: Core & Shell (1-5 points)
- ▶ BD+C: Schools (1-5 points)
- ▶ BD+C: Retail (1-5 points)
- ▶ BD+C: Data Centers (1-5 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-5 points)
- ▶ BD+C: Hospitality (1-5 points)
- ▶ BD+C: Healthcare (1-5 points)

Intent

To reduce the environmental and economic harms associated with fossil fuel energy and reduce greenhouse gas emissions by increasing self-supply of renewable energy and the use of grid-source, renewable energy technologies and carbon mitigation projects.

Requirements

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Use on-site renewable energy systems, procure renewable energy offsite, or purchase Energy Attribute Certificates (EACs) or carbon offsets to meet or offset annual building greenhouse gas emissions.

Projects may choose one or more strategies for renewables procurement from the categories below. Points achieved in each category may be added for up to a total of 5 points.

Environmental benefits of all procurement must be retained by the project.

All off-site qualifying resources must be contracted, owned, or leased for at least 15 years.

Existing off-site renewables, EACs and carbon offsets must be procured from projects that have come online or been built within the last 15 years.

Table 1. Points for Renewables Procurement

Points	On-Site Renewables	New Off-Site Renewables	Existing Off-Site Renewables	Green-e Certified: EACs & Carbon Offsets	EACs & Carbon Offsets
1	2%	20%	60%	100%	150%
2	6%	40%	80%	200%	
3	15%	60%	100%	300%	
4	35%	80%			
5	60%	100%			
EP	100%				

New Off-site Renewables: built within the last year or contracted prior to renewable energy project development.

Existing Off-site Renewables: contracted from an existing renewable energy provider.

Guidance

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Renewable energy criteria in v4.1 has been expanded to recognize the variety of procurement strategies that help to add renewable energy to the grid. Renewable energy generation can translate to greenhouse gas emission reductions and offer local environmental benefits by reducing air pollution and increasing resilience. Renewable energy produced on-site protects projects from energy price volatility while reducing wasted energy lost in transmission.

Additionally, the voluntary market can be an effective catalyst for encouraging energy generators and utility companies to develop clean energy sources and help address climate change.

Purchasing energy attribute certificates (EACs) allows buildings that use nonrenewable power to create market demand for renewable energy and support the development of renewable infrastructure.

Beta Update

This new credit combines Green Power and Renewable Energy Production into one credit, recognizing the wide spectrum of renewable energy procurement. The credit adds new categories of renewables and updates performance requirements. The credit structure incentivizes self-supply of renewable energy, development of new renewables, and real carbon emissions reductions, and provides further opportunities for building and portfolio owners to select the renewable procurement strategies that are most appropriate for the project application.

Step-by-Step Guidance

Step 1. Explore opportunities for renewables procurement.

(See Further Explanation, *Renewable Resource Procurement options*)

Step 2. Compare requirements for renewable energy systems and off-site methods of procurement.

- ▶ Carefully evaluate the space requirements (for on-site systems), costs, financial incentives, and efficiencies for each potential renewable technology or contract.
- ▶ Local funding, financing, and incentives for renewable generation projects may be available for certain technologies and may be a significant factor. When considering funding options, ensure that the terms of the contract will address all renewable attributes to be retained by the project.
- ▶ For on-site systems, excess energy, beyond the building's energy demand at a given point, can be sold to the utility company (net metering). The building owner receives the market rate, however, and cannot charge a premium for the renewable energy. In effect, the grid serves as a storage system and frees the project from hosting a storage system on site. Alternatively, project teams may consider including a storage system to increase resiliency and facilitate further control of building energy costs.
- ▶ Tying into an existing community system or creating a community system may lower cost barriers through economies of scale, because unit costs may decrease as system sizes increase. Community systems can also take advantage of time-shifted demand: one building that is occupied during the day and another building that is occupied at night could both take advantage of the same biofuel-fired heating system.
- ▶ For buildings that are part of a portfolio of buildings,

- ▶ Renewable energy may be available from a third-party system, or the project team may enter an arrangement in which a third party owns a system that serves the project. In such cases, project teams must take additional steps to ensure that the arrangement continues for the contract period required in the credit, and that the renewable attributes are retained throughout the duration of the contract

Undertake a cost-benefit analysis to understand the financial and environmental benefit of all available options.

Step 3. Set target for renewable energy procurement.

Select one or more procurement strategies, for a total of up to 5 points. Each procurement strategy must meet or exceed the minimum target for offsetting annual building greenhouse gas emissions specified in the credit language.

To establish the target renewable energy system size for the project, estimate the annual greenhouse gas emissions for the project.

Projects that use the Energy Performance Compliance Path to achieve EA Credit Optimize Energy Performance (Option 1) must base annual greenhouse gas emissions on the whole-building simulation results for the Proposed Building Performance model (see *Further Explanation, Example 1*).

Projects that use the prescriptive path to achieve EA Prerequisite Minimum Energy Performance must use the U.S. Department of Energy's Commercial Buildings Energy Consumption Survey (CBECS) data to estimate annual energy use (see *Further Explanation, Example 2*).

Review credit point thresholds and establish the renewable procurement goals for the project.

Step 4. Finalize Renewable Energy Procurement.

Purchase and install the renewable energy systems, and/or finalize the contract to procure renewable energy, EACs, and/or carbon offsets. Assure that the contract is signed by both parties, and that the contract terms confirm all credit requirements. For on-site and off-site renewable energy system(s), review the contract to confirm that the renewable system(s) are scheduled to be operational at the time of building occupancy. On-site renewable systems must also be commissioned per the requirements of LEED EA prerequisite Fundamental Commissioning and Verification and EA credit Enhanced Commissioning, as applicable.

Further Explanation

Calculations

Total building annual greenhouse gas emissions

Use Equation 1 to estimate the total building annual greenhouse gas emissions. Projects that use whole-building simulation to comply with EA Credit Optimize Energy Performance must use the results of the model in this calculation. Projects that use the prescriptive path to achieve EA Credit Optimize Energy Performance (Option 2 or 3) must use the U.S. Department of Energy's Commercial Buildings Energy Consumption Survey (CBECS) data to estimate annual energy use.

Equation 1: Total annual greenhouse gas emissions

$$CO2e_T = \sum_i \text{EnergySource}_i \times CO2e\text{Coeff}_i$$

Where:

- $CO2e_T$ is the total building annual greenhouse gas emissions before accounting for renewable resources
- $EnergySource_i$ is the annual energy consumption of each energy source serving the project (e.g. kWh/year Electricity or therms/year of Natural Gas)
- $CO2eCoeff_i$ is the CO_2 equivalent emissions coefficient per unit of energy generated, consistent with the coefficients from EA Prerequisite Minimum Energy Performance (e.g. metric tons of CO_{2e} per kWh for regional grid electricity, or metric tons of CO_{2e} per therm of natural gas)

Use Equation 2 to estimate the annual greenhouse gas emissions offset by renewable energy for on-site and off-site renewables with zero carbon emissions. Apply the equation separately to calculate greenhouse gas emissions offset for on-site renewable energy, new off-site renewable energy, existing off-site renewable energy, green-e certified EACs and carbon off-sites, and other EACs and carbon offsets as applicable:

Equation 2: Annual greenhouse gas emissions offset by renewable energy, EACs, and carbon offsets

$$CO2e_R = \sum_i RenewableSource_i \times CO2eCoeff_i$$

Where:

- $CO2e_R$ is the annual greenhouse gas emissions offset by renewable energy
- $RenewableSource_i$ is the annual energy offset by each renewable source serving the project (e.g. kWh/year Electricity)
- $CO2eCoeff_i$ is the CO_2 equivalent emissions coefficient per unit of energy generated for the energy source replaced by the renewable energy, consistent with the coefficients from EA Prerequisite Minimum Energy Performance (e.g. metric tons of CO_{2e} per kWh for regional grid electricity, or metric tons of CO_{2e} per therm of natural gas)

For renewable energy systems with some carbon emissions (such as biofuel), contact USGBC to discuss the calculation methodology.

Use Equation 3 to estimate the annual percentage greenhouse gas emissions offset through renewables. Apply the equation separately to calculate percentage greenhouse gas emissions offset for on-site renewable energy, new off-site renewable energy, and existing off-site renewable energy as applicable.

Equation 3. Percentage greenhouse gas emissions offset through renewables

% greenhouse gas emissions offset through renewables = $CO2e_R / CO2e_T$

Equation 4. Percent emission reduction from EACs and Carbon Offsets

$$CO2e_{R,EACs} / (CO2e_T - CO2e_{R,Onsite} - CO2e_{R,NewOffsite} - CO2e_{R,ExistingOffsite})$$

For projects with a contract length of less than fifteen years, use Equation 5 to prorate the annual renewable energy purchase, annual EAC purchase, and/or carbon offset purchase:

Equation 5: Proration of off-site renewables, EACs, and Carbon Offsets

$$\text{Equivalent Annual Purchase Amount} = \text{Annual Purchase Amount} \times \text{Contract Length} / 15$$

For bulk purchase, length of contract should be entered as 1 year.

Sum of points from multiple procurement sources

Using Table 1, add up the applicable points from equation 2, for on-site renewables and new and existing off-site renewables; and equation 4 for EACs and carbon offsets (either Green-e or other), for a total not to exceed 5 points plus 1 additional exemplary performance point if 100% of emissions are offset using on-site renewable sources.

Renewable Energy Considerations

The renewable energy credit seeks to increase overall demand for renewable energy and the use of grid-source renewable energy and carbon mitigation projects, with the goal of supporting broader grid-scale decarbonization. Criteria rewards renewable energy investments that have a high probability of causality (i.e. support development and installation of new renewables) and demonstrate long-term commitment. Project teams should follow a hierarchy for selecting renewable energy sources to meet credit requirements:

- ▶ First, on-site generation;
- ▶ second, local generation, such as community solar or wind, in instances where it will have a beneficial decarbonizing impact;
- ▶ third, offsite generation projects with high probability of causality, e.g. power purchase agreements;
- ▶ fourth, renewables from an existing renewable energy project, e.g. utility green tariff or direct access to wholesale markets
- ▶ last, energy attribute certificates (EACs) and/or carbon offsets

The U.S. EPA's [Guide to Purchasing Green Power](#) provides additional information on the process of and strategies for procuring renewable energy.

On-Site Renewable Energy System Considerations

On-site renewable energy generation, when combined with careful consideration of building energy time of use and grid peak demand, and storage in some grid regions, can reduce annual greenhouse gas emissions, increase building resilience, and support effective grid management.

To qualify as an on-site system, the renewable energy must be generated on-site from renewable sources produced at the building or contiguous campus site. Examples of on-site renewable energy generation include:

- ▶ A photovoltaic array located on the project site.
- ▶ A wind tower located on a contiguous campus owned by the same entity as the project building.
- ▶ Landfill gas processed in digesters on a contiguous campus owned by the same entity as the project building, and used to produce thermal energy in the project building.

Note: earlier versions of LEED allowed some biofuels produced off-site to qualify as on-site renewable energy. However, based on the clarifications provided in ASHRAE 90.1-2016 for on-site renewable energy, and the clearer distinction between on-site and off-site renewable energy in LEED v4.1, biofuels are only considered on-site renewable systems when the renewable source is harvested on site, and used for on-site generation of electric or thermal energy.

Only usable energy generated from the on-site renewable system shall be considered towards the on-site renewable energy contribution. Usable energy is defined as the output energy from the system less any transmission and conversion losses, such as standby heat loss, losses when converting electricity from DC to AC, or waste heat in a cogeneration system that is exhausted to the atmosphere during periods of low thermal demand. Excess energy, beyond the building's energy demand at a given point, can be sold to the utility company (net metering).

A project team should use web resources and other tools available to determine the feasibility of renewable systems, given the project site's climate, context, and infrastructure. Consider the features of the site, such as solar availability, wind patterns, and other renewable energy sources, and any seasonal or daily variations in its supply. Certain project types may have special opportunities: office or university campuses typically have available land, for example, and warehouse projects may have large roof areas.

Match the project's energy needs with renewable energy output when selecting a renewable system. For example, a sunny site is a good candidate for solar thermal hot water, but this type of renewable resource is most cost-effective if the building has a constant demand for hot water. Accordingly, a hotel or a multifamily project may be a better match for a solar thermal hot water system than an office complex.

Daily and seasonal variations in loads also factor into the investigation of renewable energy. For example, a residential project with low daytime electricity demand may require battery storage to benefit from a photovoltaic (PV) array; an office building with high daytime demand may not.

On-site renewable systems must be installed and commissioned prior to LEED construction phase project submission to qualify for on-site renewable energy generation credit.

New Off-Site Renewables

New off-site renewables are defined as those that have come online within the last year or contracted for the building project (or for a portfolio of building projects or tenant improvement spaces) prior to the development of the renewable energy project in which the project is investing.

The renewable energy system must be operational at the time of LEED project construction phase submittal to earn credit.

Community renewable energy cooperatives, larger-scale investments, such as direct, voluntary purchases in the form of power purchase agreements (PPAs), virtual PPAs, or renewable energy investment trusts, qualify as new off-site renewables provided documentation demonstrates that they meet the criteria described above. Contracts for these investments must indicate the specific system used to generate the renewable energy, with sufficient information available to confirm the renewable system generation capacity and allocation of the EACs (see Further Explanation, Renewable Attributes).

Investment in new off-site renewables creates new renewable energy supply and displaces energy and emissions from fossil fuel-powered generators, particularly in regions where the grid mix is a higher percentage of fossil fuels.

To qualify as a new renewable system, the contract length shall be a minimum of fifteen years, or the annual energy renewable energy generation shall be prorated based on the contract term length. A commitment to renew does not qualify as a new renewable resource.

Existing Off-Site Renewables

Existing off-site renewables are defined as those contracted from an existing renewable energy provider or off-site renewable systems that were contracted for the building after the renewable system came online and came online more than one year before building occupancy.

Existing off-site renewables, which include utility green tariff programs or direct access to wholesale markets, may be more widely available depending on project location or budget. Investment in existing renewable resources and utility programs remains an important strategy for sustaining market demand for renewables and ensuring financial viability of existing projects.

The contract length shall be a minimum of fifteen years, or the annual energy renewable energy generation shall be prorated based on the contract term length. Alternatively, for utility green tariff programs where a fifteen-year contract is not available, project teams may show compliance with the fifteen year minimum contract term by demonstrating that the project has an executed contract for the maximum contract length available (with a minimum of one year), and the building owner must provide a signed letter of commitment indicating that the project will continue to renew the renewable contract or engage in an alternate renewable energy contract for a total of fifteen years.

The contract for existing renewable contracts shall confirm that all contributing renewable resources have come online or been built within the last fifteen years to qualify for credit compliance.

Energy Attribute Certificates and Carbon Offsets

The contract for EACs and carbon offsets shall confirm that all contributing renewable resources or carbon offsets are from projects that have come online or been built within the last fifteen years to qualify for credit compliance.

To earn up to three points under the credit, projects may purchase EACs and carbon offsets meeting the Green-e Renewable Energy Standard or the Green-e Climate Standard with an annual purchase amount that offsets 100% to 300% of total annual greenhouse gas emissions for a total of fifteen years.

Projects not using Green-e certified products can earn one point under the credit by purchasing annual EACs and carbon offsets that offset 150% of total annual greenhouse gas emissions for a total of fifteen years. These must be certified under an eco-label developed by an independent organization with transparent accounting process and standards in place which address the following:

- ▶ Verifiable chain of custody
- ▶ Verifiable age of renewable energy
- ▶ Tracking of GHG reductions from eligible projects
- ▶ Mechanism to prevent double-counting
- ▶ Third party-verified retail transaction

The executed contract must specify the purchasing goals and is valid for the duration indicated in the credit requirements, or includes a bulk purchase amount with the total purchase amount matching the total annual amount indicated in the credit requirements multiplied by fifteen years.

All of the above procurement strategies play an important role in reducing global greenhouse gas emissions. Leveraging a combination of procurement strategies can send important demand signals and spur further investment in renewables markets globally.

Required Documentation

Documentation	On-site system	Off-site Renewables
Renewable system rated capacity	X	X
Calculations to determine energy generated	X	X
Calculations to determine greenhouse gas emissions avoided from renewable generation or off-site procurement	X	X
Documentation of annual greenhouse gas emissions	X	X

Confirmation of renewable attribute ownership	X (if owned by building owner)	
Contract indicating duration and renewable attribute ownership	X (if owned by 3 rd party)	X
Contract indicating percentage ownership, lease, or allocation of new off-site renewable system, and specific location of new off-site system		X (New)
If selling EACs for on-site system, contract and Green-e certification for EAC or carbon offset purchase for a minimum of fifteen years	X	
Calculations showing renewable electricity, EACs, or carbon offsets for targeted point threshold		X
Purchase letter or contract of commitment showing renewable electricity, EACs, or carbon offsets for targeted point threshold and confirmation of the age of system		X
Eco-label documentation showing label development, transparent accounting processes and standards, if not Green-e certified		X

Connection to Ongoing Building Performance

- LEED O+M EA credit Energy Performance: Investments in renewable energy throughout the building life cycle can help reduce building greenhouse gas emissions and improve the building's energy performance score, increase market demand for renewables, and support the growth and financial feasibility of new renewable energy projects.